

Electrical magnetohydrodynamic flow of kerosene oil based carbon nanotube's Maxwell nanofluid in the presence of non-linear radiation and Cattaneo-Christov heat diffusion: Applications in aerospace industry

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Sangapatnam Suneetha¹, Lal Ahamed Wahidunnisa¹,
Abbai Reddy Divya² and Polu Bala Anki Reddy²

Abstract

This paper investigates the influence of carbon nanotubes to improve heat transmission with Maxwell nanofluids which is suspended in kerosene oil. This mathematical model has been expressed in terms of partial differential equations, which are transformed by using compatible variables into ordinary differential equation's. The result of this model has been attained numerically via RK fourth order with shooting iteration. The results show that with a larger magnetic field, heat transfer increases radiation while decreasing velocity. The combined electrical magnetohydrodynamic effects have the potential to improve fluid flow rates. Moreover, the non-linear radiation case is dominating to the linear case. This theoretical study may be applied in material science, purification technologies, optics and electricity owing to their astonishing possessions, aerospace industry and many more. Especially in aerospace, by way of the aid in plummeting the weight, satellite weight lessening, lightning protection for aircraft and so on.

Keywords

Electrical magnetohydrodynamic, carbon nanotube's, non-linear thermal radiation, Cattaneo-Christov heat diffusion, Maxwell nanofluid

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Introduction

It is evident that all the bodies will transmit heat, and greater will the temperature thereby producing more noticeable radiation, to the point that, when two bodies of innumerable temperatures are inside one other, the hotter body would transfer heat towards the colder body till they reach an alike temperature.¹ Prevost's Theory of Exchange is a logical declaration that is frequently referred to as the initial point for the cutting-edge real concept of thermal radiation in the nineteenth century.² The parametrization and non-dimensionality of the thermal radiation archetypal have become a modest chore. The electromagnetic radiation that emerges in the fluid, while it flows for a portion of the mechanical cycles might not be useful if deciphered as direct. Furthermore, the Taylor series extension could not capture all the impact's intricacies. Non-linear thermal radiation has been recently presented as a model for precise clarification considering this. Reddy³ explains combustion chambers, re-entry aerothermodynamics, aircraft, propulsion devices, un-oriented

power innovation, furnace operations and, chemical processes at high temperatures, as examples of where nonlinear thermal radiation can be exploited. We can take nonlinear thermal radiation into account because of these challenges. In a related study, Animasaun⁴ looked into unstable magnetohydrodynamic and micropolar liquid flow past an upward permeable plate passing through a double combination in an optically feeble environment. In the manifestation of Joule heating over a stretched sheet, the sway of non-linear thermal radiation on the three-dimensional (3D) growth of Jeffrey nanofluid was examined by Kumar

¹Department of Mathematics, Yogi Vemana University, Kadapa, Andhra Pradesh, India

²Department of Mathematics, S.A.S., Vellore Institute of Technology (VIT), Vellore, India

Corresponding author:

P. Bala Anki Reddy, Department of Mathematics, S.A.S., Vellore Institute of Technology (VIT), Vellore-632014, India.
Email: pbarmaths@gmail.com

et al.⁵ Polu and Reddy⁶ investigated the effect of heat radiation combined with viscous dissipation happening magnetohydrodynamic unsteady flow through an exponentially inclined preamble stretching sheet.

The thermo-physical and volume portioning methods of nanoparticles are used to determine the heat transfer properties of non-materials. Carbides, oxides, nitrides, fullerene, metals, and carbon nanotubes (CNTs) are among the numerous types of nano fluids based on their size, shape, and characteristics.⁷ We've been looking into CNTs because of their enhanced heat conductivity. CNTs have a chambered nanostructure and are comprised of allotropes of carbon. The physiognomies of carbon particles in round and hollow structures are interesting and useful in health sciences, applied sciences, texture sciences, energy, optics, real sciences and assembly.⁸ CNTs are divided into two types namely, single-walled CNT (SWCNT) and multi-walled CNT (MWCNT) based on their form. In 1991, Iijima⁹ was the first to utilize terminology with reference to CNTs when he investigated MWCNTs employing the Krastschmer and Huffman technique. Ajayan and Iijima¹⁰ accounted for SWCNTs 1993. Nanofluids are predicted to alleviate the problems that come with materials that have low thermal conductivities, such as water, motor oil, coal oil and fuel. Areekara et al.¹¹ investigated how water flowing through SWCNTs with magnetite nanoparticles on the stagnation point flows over a stretching sheet using a bio-convective model.

Over the last two centuries, Fourier's law of thermal conduction has been used to inspect the qualities of heat transport. In any case, this concept is unsatisfactory since any underlying aggravation is felt across the entire substance immediately.¹² To resolve this concern, Cattaneo inserted a thermal relaxation period in the classic Fourier's law of heat conduction, which allowed for limited heat transfer via thermal waves. After that, alongside Oldroyd's upper-connected subsidiaries, Christov altered the Cattaneo law by thermal relaxation time to get the material-invariant definition. Gireesha et al.¹³ employed a Cattaneo–Christov thermal flux-based enhanced sort of Fourier's law. The elements of magnetohydrodynamics flow through melting heat transfer of dusty Casson liquid over an extended sheet have been demonstrated to make advantage of this. Nayak et al.¹⁴ investigated the Cattaneo–Christov dispersion concept through heat mass transitions with two-dimensional (2D) incompressible progression of a micropolar mixed nanofluid over a straight extending/contracting sheet. Shankaralingappa et al.^{15–18} address the Cattaneo–Christov transmission of heat of a dusty tangent hyperbolic liquid over an extending sheet.

Svante Arrhenius initially coined the term activation energy in 1889.¹⁹ Also, activation energy defined as the energy that is recovered to be defeated for a drug response. It also has the lowest energy required to initiate a synthetic reaction, such as a chemical reaction. Medications, thermal powers, and emulsions of oil and water are all examples of activation energy applications. Abbas et al.²⁰ considered the impacts of binary chemical reactions thru Arrhenius activation energy by evaluating chemical reaction rate as a function of temperature. Reddy et al.¹⁸ probed the combined

effects of Arrhenius initiation energy, synthetic response, with heat generation/absorption on a 3D magnetohydrodynamic stream over a slendering stretchy sheet. Shamshuddin and Mabood²¹ probed the flow of a micropolar nanofluid through suspended nanoparticles across a porous extending sheet thru non-uniform heat source/sink, binary chemical reaction with activation energy.

Based on the above concepts, the purpose of this research is to undertake a theoretical analysis of an EMHD viscous fluid with CNTs across a stretching sheet thru the occurrence of non-linear radiation formation. We used the suitable similarity transmutations to convert the leading equations to ordinary differential equations (ODEs) and then used the RK fourth order shooting technique to resolve the problems. We also looked at the friction factor and heat transfer rate by tabulating their results against the same factors that we have graphs for.

Mathematical formulation

We investigate the situation of continuous 2D EMHD flow of an incompressible Newtonian fluid flow in the incidence of nanofluids like SWCNTs and MWCNTs thru kerosene by way of a base fluid. The electric with magnetic field stimuluses, which are studied on nanofluid, follow Ohm's law. Furthermore, with the variety of nanofluids, the activation energy source term and nonlinear thermal radiation effect have also been designated. The following are the governing equations^{22–24}:

$$u_x + v_y = 0 \quad (1)$$

$$\begin{aligned} & uu_x + vv_y + \lambda_1(u^2u_{xx} + v^2u_{yy} + 2uvu_{xy}) \\ &= \frac{1}{\rho_{nf}}(\mu_{nf}u_{yy} + g((\rho\beta)_f(T - T_\infty) - (\rho\beta^*)_f \\ &\quad \times (C - C_\infty)) + \sigma_{nf}BE_0 - \sigma_{nf}B^2u) \end{aligned} \quad (2)$$

$$\begin{aligned} (\rho C_p)_{nf}(uT_x + vT_y) &= -\nabla \cdot q - \frac{1}{v}(q_r)_y + \mu(u_y)^2 \\ &+ Q_0(T - T_\infty) \end{aligned} \quad (3)$$

$$uC_x + vC_y = DC_{yy} - \frac{k_r^2}{\left(\frac{T}{T_\infty}\right)^{-m}}(C - C_\infty)e^{-\frac{E_a}{kT}} \quad (4)$$

In equation (4), the term $k_r^2\left(\frac{T}{T_\infty}\right)^m e^{-\frac{E_a}{kT}}$, E_a signifies the modified Arrhenius equation, and activation energy respectively, m (will be $1 > m > -1$) be the fixed rate constants, k_r^2 stand for the reaction rate, and Boltzmann constant k which is equal to 8.61×10^{-5} eV / K (Figure 1).

The Cattaneo–Christov heat flux is conveyed by way of:

$$\begin{aligned} & q + \lambda_2 q_t + \lambda_2 V \cdot \nabla q - \lambda_2 q \cdot \nabla V + \lambda_2 (\nabla \cdot V)q \\ &= -k_f \nabla \cdot T \end{aligned} \quad (5)$$

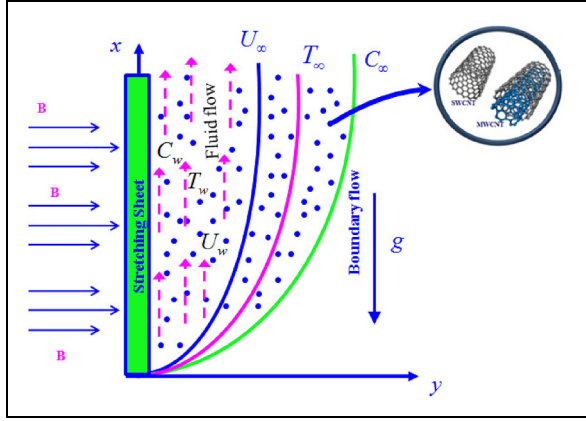


Figure 1. Flow outline of the problem.

Here, λ_2 , k_f , V signifies the relaxation time of heat flux, thermal conductivity of the prime fluid, and velocity vector, respectively.

Suppose, λ_2 is equal to zero then equation (5) is transformed into Fourier's law and the fluid, which is being incompressible, $\nabla \cdot V$ is equal to zero, and hence equation (5) receipts the form as:

$$q + \lambda_2 q_t + \lambda_2 V \cdot \nabla q - \lambda_2 q \cdot \nabla V + k_f \nabla \cdot T = 0 \quad (6)$$

After jettisoning q from equations (3) and (6), the energy equation for the case of steady flow as:

$$\begin{aligned} uT_x + \lambda_2((uu_x + vu_y)T_x + (uv_x + vv_y)T_y \\ + (uT_x + vT_y)^2) + vT_y = \alpha_{nf} T_{yy} - \frac{1}{v} (q_r)_y + \frac{v}{c_p} (u_y)^2 \end{aligned} \quad (7)$$

By using Rosseland's approximation, the radiation heat flux q_r is labeled as:

$$q_r = -4 \left(\frac{1}{3K^*} \right) \text{grad}(e_b) \quad (8)$$

absorption coefficient, blackbody emission power, absolute temperature, Stefan-Boltzmann radiation law and is the Stefan-Boltzmann constant, respectively.

The term T^4 may possibly be stated as a first order function of temperature itself and so T^4 may possibly be elongated by way of Taylor series in terms of T_∞ and estimated subsequently eradicating higher order terms.

$$T^4 = \frac{\{1 + (\theta_w - 1)\theta\}^4}{T_\infty^{-4}} \quad (9)$$

where $\theta_w = \frac{T_w}{T_\infty}$, $\theta_w > 1$ designates as the wall temperature of the ratio parameter.

The thermal conductivities of CNTs are prophesied by facsimilia for instance Maxwell, Jeffery, Davis, Crosser, and Hamilton. These models, according to Xue, are designed for spherical or elliptical particles thru a slight axial ratio. The inspiration of CNT space distribution on heat conductivity is not buoyed by these simulations. As a result, he developed a model established on Maxwell's theory for rotational elliptical nanotubes thru a hefty axial ratio that compensates for the influence of spatial dissemination on CNTs. The following is how the Xue exemplary is used in this paper:

$$\frac{k_{nf}}{k_f} = \frac{1}{\frac{2\chi}{\left(\frac{k_{CNT} - k_f}{k_f}\right)} \ln \frac{k_{CNT} + k_f}{2k_f} + 1 - \chi} \quad (10)$$

which are interpreted as:

$$\left. \begin{aligned} v_{nf} &= \frac{1}{\rho_{nf}}, \quad \mu_{nf} = \mu_f \frac{1}{(1 - \chi)^{2.5}}, \quad \rho_{nf} = \rho_f - \chi \rho_f + \chi \rho_{CNT}, \\ \sigma_{nf} \frac{1}{\sigma_f} &= 1 + \frac{3 \left(\frac{\sigma_{cnt}}{\sigma_f} - 1 \right) \chi}{\left(\frac{\sigma_{cnt}}{\sigma_f} + 2 \right) - \left(\frac{\sigma_{cnt}}{\sigma_f} - 1 \right) \chi}, \\ \alpha_{nf} &= \frac{k_{nf}}{(\rho C_p)_{nf}}, \quad (\rho C_p)_{nf} = (1 - \chi)(\rho C_p)_f + \chi(\rho C_p)_{CNT} \end{aligned} \right\} \quad (11)$$

Here $K^* \left(\frac{1}{m} \right)$, $e_b \left(\frac{W}{m^2} \right)$, T , $e_b = \frac{\sigma^*}{T^4}$, $\sigma^* = 5.6697 \times 10^{-8} \frac{W}{m^2 K^4}$ are exemplifies the Rosseland mean spectral

A boundary layer is a layer of fluid that develops near a surface that separates the fluid from it. Every time

a fluid travels over a surface, we have a boundary layer at the surface.²⁵ Theme to the periphery circumstances:

$$\begin{aligned} u &= U_w(x) + A \frac{\partial u}{\partial y}, \quad \nu = -v_w, \quad T = T_w + B \frac{\partial T}{\partial y}, \\ C &= C_w + D \frac{\partial C}{\partial y} \quad \text{at } y = 0 \\ u &\rightarrow 0, \quad T \rightarrow T_\infty, \quad C \rightarrow C_\infty \quad \text{as } y \rightarrow \infty \end{aligned} \quad (12)$$

Using the appropriate similarity conversions,

$$\begin{aligned} \eta &= \frac{y}{\sqrt{\frac{v}{c}}}, \quad \psi(x, y) = \sqrt{cv}xf(\eta), \\ \theta(\eta) &= \frac{T - T_\infty}{T_w - T_\infty}, \quad \phi(\eta) = \frac{C - C_\infty}{C_w - C_\infty}, \\ u &= cx f'(\eta), \quad \nu = -\sqrt{cv}f(\eta) \end{aligned} \quad (13)$$

Implementing equations. (10), (11), (13) into equations (1), (2), (7), (4), we acquire

where

$$\begin{aligned} Sc &= \frac{v}{D}, \quad \sigma = \frac{k_r^2}{c}, \quad E = \frac{E_a}{kT_\infty}, \quad \lambda = \frac{Gr_x}{Re_x^2}, \\ Gr_x &= \frac{(\rho\beta)_f g(T_w - T_\infty)x^3}{v^2 \rho_f}, \quad Nr = \frac{Gc_x}{Re_x^2}, \\ Gc_x &= \frac{(\rho\beta^*)_f g(C_w - C_\infty)x^3}{v^2 \rho_f}, \quad Re_x^2 = \frac{U_w^2 x^2}{v^2}, \\ M &= B \sqrt{\frac{\sigma}{a\rho_f}} x, \quad \alpha = \lambda_1 c, \quad \gamma = \lambda_2 c, \\ Rd &= \frac{4\sigma^* T_\infty^3}{kk_1}, \quad Ec = U_w^2 \frac{1}{C_p(T_w - T_\infty)}, \\ pr &= \frac{(\mu C_p)_f}{k}, \quad S = v_w \frac{1}{av_f}, \\ \delta &= A \sqrt{\frac{c}{v}}, \quad Bi = \left(\frac{h_f}{k_f}\right) \sqrt{\frac{v_f}{c}}. \end{aligned} \quad (18)$$

$$\begin{aligned} f''' + (1 - \chi)^{2.5} \left\{ \left(1 - \chi + \chi \frac{(\rho\beta)_{CNT}}{(\rho\beta)_f} \right) (\lambda\theta - Nr\phi) - \left(1 + \frac{3 \left(\frac{\sigma_{cnt}}{\sigma_f} - 1 \right) \chi}{\left(\frac{\sigma_{cnt}}{\sigma_f} + 2 \right) - \left(\frac{\sigma_{cnt}}{\sigma_f} - 1 \right) \chi} \right) (Mf' - EI) \right\} \\ - (1 - \chi)^{2.5} \left(1 - \chi + \chi \frac{\rho_{CNT}}{\rho_f} \right) \{ f'^2 - f f'' - 2\alpha f f' f'' + \alpha f^2 f''' \} = 0 \end{aligned} \quad (14)$$

$$\begin{aligned} \frac{1}{pr} \left\{ \theta'' \left\{ \frac{k_{nf}}{k_f} + \frac{4}{3} Rd[(\theta_w - 1)\theta + 1]^3 \right\} + 4Rd[(\theta_w - 1)\theta + 1]^2 (\theta_w - 1)\theta'^2 \right\} \\ + \left\{ 1 - \chi + \chi \frac{(\rho c_p)_{CNT}}{(\rho c_p)_f} \right\} \{ f \theta' - \gamma(f f' \theta' + f^2 \theta'') \} + Ec f''^2 + Q\theta = 0 \end{aligned} \quad (15)$$

$$\phi'' + Scf \phi' - \frac{Sc \sigma \phi [\theta(\theta_w - 1) + 1]^m}{e^{\theta(\theta_w - \theta) + 1}} = 0 \quad (16)$$

alongside through the periphery circumstances

$$\begin{aligned} f(\eta) &= S, \quad f'(\eta) = 1 + f''(0)v_s, \quad \theta(\eta) = 1 + \theta'(0)t_s, \\ \phi(\eta) &= 1 + \phi'(0)c_s \quad \text{at } \eta = 0 \\ f'(\eta) &= 0, \quad \theta(\eta) = 0, \quad \phi(\eta) = 0 \quad \text{at } \eta = 0 \text{ at } \eta \rightarrow \infty \end{aligned} \quad (17)$$

Quantities of interest

The surface drag force $C_{fx} = \frac{\tau_w}{\rho_f U_w^2}$, Nusselt and Sherwood number are $Nu_x = \frac{xq_w}{k_f(T_w - T_\infty)}$, $Sh_x = \frac{xq_m}{D(C_w - C_\infty)}$ correspondingly the quantities of interest when considering flow:

where $\tau_w = \mu_{nf} \left(\frac{\partial u}{\partial y} \right)_{y=0}$ (wall skin friction), $q_w = -xk_{nf} \left(\frac{\partial T}{\partial y} \right)_{y=0} + (q_r)_{y=0}$ (wall heat flux) and $q_m = -D \left(\frac{\partial C}{\partial y} \right)_{y=0}$ (wall mass flux)

From equations (14) and (19), the non-dimensional surface drag force $C_f Re_x^{1/2} = \frac{1}{(1-\chi)^{2.5}} f''(0)$, heat and mass

Table 1. Thermo physical selves of CNTs through base fluid^{26–28}.

Physical possessions	Nano particles		Fluid phase (Kerosene)
	SWCNTs	MWCNTs	
C_p (J/kg K)	425	2090	796
ρ (kg/m ³)	2600	783	1600
κ (W/m K)	6600	0.15	3000
$\beta \times 10^{-5}$ (K ⁻¹)	2.6	2.1×10^{-6}	2.8

CNT: carbon nanotube; SWCNT: single-walled CNT; MWCNT: multi-walled CNT.

transfer rates are $Nu_x Re_x^{-\frac{1}{2}} = -\left\{\frac{k_{nf}}{k_f} + \frac{4}{3} Rd[(\theta_w - 1)\theta(0) + 1]\right\} \theta'(0)$ and $Sh_x Re_x^{-\frac{1}{2}} = -\phi'(0)$ respectively.

Currently $Re_x = \frac{U_w x}{\nu}$ represents the local Reynolds number (Table 1).

Numerical procedure

To get numerical solutions to coupled ordinary equations with boundary conditions, the Runge-Kutta Fourth order thru shooting iteration is exploited. The following procedure is used to convert a system of linked non-linear ODEs into a system of first order ODE's:

$$\begin{aligned} f &= D_1, \quad f' = D_2, \quad f'' = D_3, \quad \theta = D_4, \\ \theta' &= D_5, \quad \phi = D_6, \quad \phi' = D_7 \end{aligned} \quad (19)$$

$$f''' = -\frac{(1-\chi)^{-2.5} D_1^{-2}}{-\left(1 - \chi + \chi \frac{\rho_{CNT}}{\rho_f}\right) \alpha} \begin{pmatrix} \left(\frac{1}{(1-\chi)^{2.5}} \left(1 - \chi + \chi \frac{(\rho\beta)_{CNT}}{(\rho\beta)_f}\right) (\lambda D_4 - Nr D_6) \right) \\ - \left(1 + \frac{3 \left(\frac{\sigma_{cnt}}{\sigma_f} - 1 \right) \chi}{\left(\frac{\sigma_{cnt}}{\sigma_f} + 2 \right) - \left(\frac{\sigma_{cnt}}{\sigma_f} - 1 \right) \chi} \right) (MD_2 - EI) \\ - \left(\frac{1}{(1-\chi)^{2.5}} \left(1 - \chi + \chi \frac{\rho_{CNT}}{\rho_f}\right) (D_2^2 - D_1 D_3 + \alpha(-2D_1 D_2 D_3)) \right) \end{pmatrix} \quad (20)$$

$$\theta'' = -\frac{1}{\frac{1}{pr} \left(\frac{k_{nf}}{k_f} + \frac{4}{3} Rd((\theta_w - 1)D_4 + 1)^3 \right)} \begin{pmatrix} \frac{1}{pr} (4Rd((\theta_w - 1)D_4 + 1)^2 (\theta_w - 1) D_5^2) \\ + \left(1 - \chi + \chi \frac{(\rho c_p)_{CNT}}{(\rho c_p)_f} \right) (D_1 D_5 - \gamma(D_1 D_2 D_5)) + Ec D_3^2 + Q D_5 \end{pmatrix} \quad (21)$$

$$\phi'' = -\left(Sc D_1 D_7 - Sc \sigma D_6 (D_4 (\theta_w - 1) + 1)^m e^{-\frac{E}{D_4 (\theta_w - 1) + 1}} \right) \quad (22)$$

The corresponding boundary conditions can be written as:

$$\begin{aligned} D_1(0) &= S, \quad D_2(0) = 1 - v_s D_3(0), \quad D_4(0) = 1 - t_s \\ D_5(0), \quad D_6(0) &= 1 - c_s D_7(0) \\ D_2(\infty) &\rightarrow 0, \quad D_4(\infty) \rightarrow 0, \quad D_6(\infty) \rightarrow 0. \end{aligned} \quad (23)$$

The values of this paper is compared with the former work by assuming some parameter values as $EI = 0$ and $Q = 0$ (Table 2).

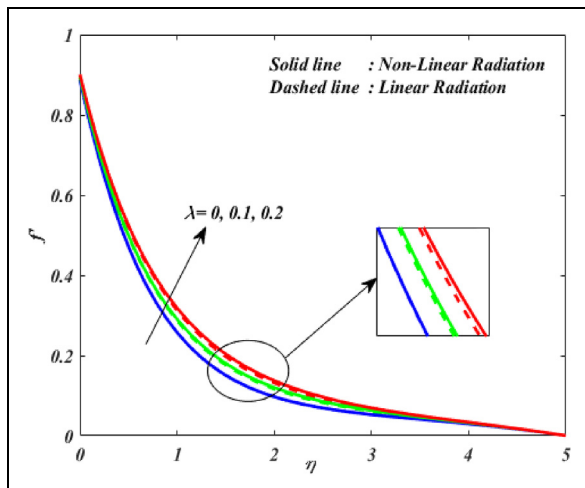
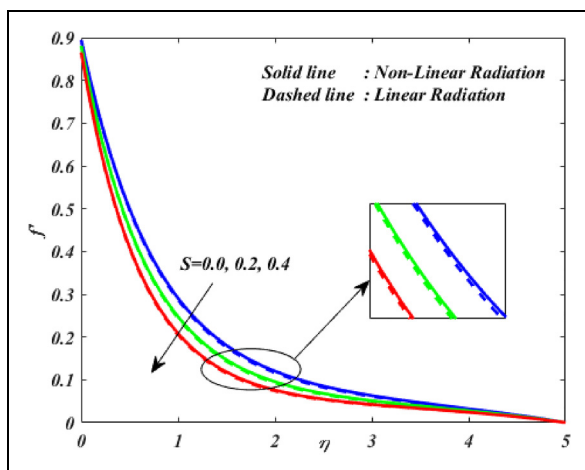
Results and discussion

The following graphs show the attitudes of different effects on velocity, temperature, and concentration. In addition to this, the impression of skin friction, the Nusselt number and Sherwood numbers have been deliberated. The stimulus of the mixed convection parameter on the velocity silhouette is exposed in Figure 2. The graph demonstrates that when the parameter is increased, the velocity rises. Figure 3 and Supplemental Figure 8 demonstrate the consequence of a suction parameter on the velocity and temperature curves. Figure 3 shows that for both linear and non-linear radiation situations, the velocity silhouette diminutions as the suction parameter rises, indicating that suction stabilizes the boundary layer. Fluid temperature decreases for the suction parameter, as shown in Supplemental Figure 8. In addition, we see that nonlinear radiation exhibits a higher temperature

Table 2. The values of skin friction coefficient and various dimensionless parameters.

M	Pr	Sc	S	$f''(0)$ Subbarayudu et al. ²³		$f''(0)$ Present	
				SWCNT	MWCNT	SWCNT	MWCNT
0.5	0.71	0.6	0.5	-0.8169	-0.7990	-0.8132	-0.7811
1.0	0.71	0.6	0.5	-0.9922	-0.9797	-0.9033	-0.9291
1.5	0.71	0.6	0.5	-1.1716	-1.1629	-1.1700	-1.1630
0.5	0.9	0.6	0.5	-0.8174	-0.7995	-0.8199	-0.7732
0.5	1.1	0.6	0.5	-0.8179	-0.8000	-0.8006	-0.7643
0.5	0.71	0.8	0.5	-0.8169	-0.7990	-0.8122	-0.7323
0.5	0.71	1.0	0.5	-0.8169	-0.7990	-0.8898	-0.7324
0.5	0.71	0.6	1.0	-0.9108	-0.8891	-0.9118	-0.8213
0.5	0.71	0.6	1.5	-0.9850	-0.9614	-0.9885	-0.9605
0.5	0.71	0.6	-0.5	-0.6067	-0.5987	-0.6090	-0.5977
0.5	0.71	0.6	-1.0	-0.5125	-0.5085	-0.5145	-0.5098
0.5	0.71	0.6	-1.5	-0.4342	-0.4326	-0.4343	-0.4334

CNT: carbon nanotube; SWCNT: single-walled CNT; MWCNT: multi-walled CNT.

**Figure 2.** Sway of λ on velocity.**Figure 3.** Sway of S on velocity.

profile than simple radiation. Figure 4 shows the prominence of the ratio of concentration buoyancy forces Nr for linear and non-linear cases on the velocity silhouette. It is notified in this figure the greater value of the Nr declines the velocity.

Figure 5 and Supplemental Figure 11 display the impression of the magnetic parameter M on velocity and temperature. When the magnetic parameter M rises, the velocity arena exhibits recessive behavior, as portrayed in Figure 5. It is physically justified that rise in magnetic parameter M , a greater Lorentz force intensity is observed. Clearly, such force acts as a barrier to fluid motion. As a result, for both linear and non-linear radiation, the velocity field decays. The Lorentz force improves the disturbance to nano liquid particles as the magnetic parameter increases, raising the temperature. For increased M , the fluid temperature simply rises. It was also discovered that in the case of nonlinear radiation, the temperature increase is greater than in the case of linear radiation.

The fluid relaxation parameter (α) is publicized on the velocity silhouette in Figure 6. As the fluid relaxation parameter is increased, the velocity drops, as illustrated in this graph (α). For viscoelastic materials, the fluid relaxation parameter, which is commonly termed as the Deborah number, is critical. Deborah's number must be expressed as a proportional quantity of distortion time.

Supplemental Figure 7 elevates the impact of velocity slip on velocity. The velocity parameter reduces with increasing velocity slip, as shown in the diagram. Furthermore, when the sheet slips, the flow velocity near the sheet and the sheet's stretching velocity change. The fluid velocity decreases when the slip criteria are met, the dragging of the sheet can partially pass through to the liquid, and the velocity gradually intensifications as the slip parameter is increased. We also discovered that in both linear and non-linear radiation instances, the velocity has the same results.

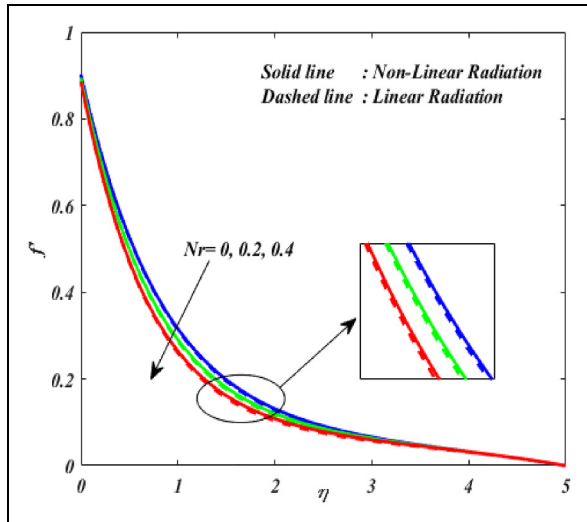


Figure 4. Sway of Nr on velocity.

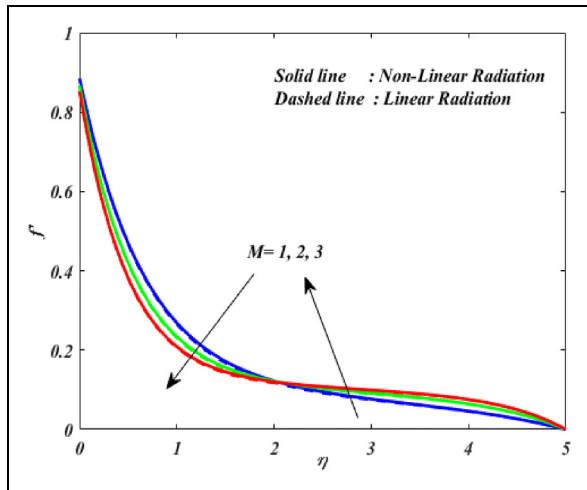


Figure 5. Sway of M on velocity.

The fluctuations in $\theta(\eta)$ for various values of the linear and non-linear thermal radiation parameter Rd are shown in Supplemental Figure 9. Temperature rises with increasing Rd which is associated boundary layer thickness, as shown in these graphs. This is because increasing the radiation parameter causes the fluid to absorb more heat, resulting in a rise in temperature and the thickness of the thermal boundary layer. Non-linear thermal radiation is more operative on temperature ($\theta(\eta)$) than linear thermal radiation, according to a critical observation.

Supplemental Figure 10 shows that as the value of Q rises, the temperature profile rises as well. When external heat is considered in the system, it increases the average kinetic energy, causing the molecules to move quickly. As a result, the fluid's temperature rises. For nonlinear radiation, the effect of the heat source parameter is more evident than for linear radiation.

The thermal relaxation parameter (γ) is depicted on the velocity parameter in Supplemental Figure 12. The velocity increases for raising values of the fluid relaxation

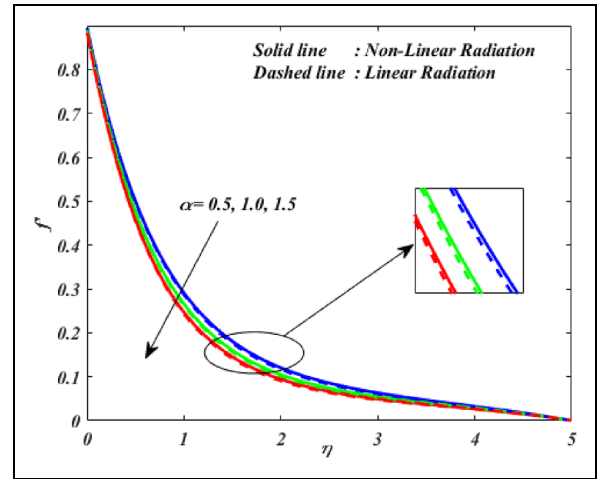


Figure 6. Sway of α on velocity.

profile, as seen in this graph (γ). When the thermal boundary layer thickness is raised, the object's particles typically take a lengthy time to transmit heat to their proximate particles. Supplemental Figure 13 spectacles the outcome of the electric field parameter $E1$, on temperature profiles for linear and nonlinear radiation scenarios.

Supplemental Figure 12 illustrations the thermal relaxation parameter (γ) on the velocity outline and it is acquainted from this diagram that the velocity silhouette intensifications for the cumulative values of the fluid relaxation parameter (γ). When the thickness of the thermal boundary layer is increased for greater values of γ , the object's particles take a long time to transfer heat to its neighbor's. Supplemental Figure 13 is sketched to scrutinize the performance of electric field parameter, $E1$ on the temperature profiles for both linear and nonlinear thermal radiation situations. The combined use of applied electric field and hybrid nanofluid can considerably improve the cooling effect of nanofluids, and the electric field can easily modify the heat transfer effect of nanofluid according to the conditions, making the cooling system operation more flexible.

As seen in Supplemental Figure 14, the fatness of the thermal boundary layer and the thermal plot increases as Ec increases. When Ec is increased, a portion of the nanofluid's mechanical energy is transformed to useful thermal energy. The temperature of the nanofluid rises because of this. The temperature profile of nonlinear radiation is higher than that of linear radiation, according to the analysis.

Supplemental Figure 15 depicted the inspiration of the thermal slip parameter going on the temperature field. It is well known that when the value of t_s lowers, the heat transfer through the surface to the next layers of fluid reduces, resulting in a lower temperature. As seen in Supplemental Figure 17, the stimulus of c_s on the mass fraction field is like that of t_s on the temperature field. It is because slip slows down fluid velocity, resulting in a reduction in net molecular movement. Basically, it slows down the fluid motion, resulting in a decrease in

net molecular mobility. As a result, decreased molecular mobility reduces both temperature and mass fraction fields, as seen in Supplemental Figures 15 and 17. Because the concentration slip parameter could be able to control the mass transport phenomena if the thermal slip parameter can control the temperature inside the flow. As a result, we investigated the flow's mixed effects of velocity, temperature and concentration slip factors. The stimulus of the concentration slip parameter is quite comparable to those of the thermal slip parameter, as we observed. Finally, the thermal and concentration boundary layers may be controlled to the appropriate values by changing the thermal and concentration slip parameters. Both linear and nonlinear radiation situations are considered. Furthermore, in the case of linear radiation, the reduction is more noticeable than in the case of nonlinear radiation. Because Sc is contrariwise associated to the mass diffusion coefficient D , bigger Sc values imply smaller D , resulting in thinner concentrations at the boundary layer. When a result, as Sc increases, the nanoparticle concentration falls for both kinds of radiation, as seen in Supplemental Figure 16. Supplemental Figure 18 demonstrates the outcome of θ_w on the concentration profile. Due to the concentration of the boundary layer thickness, the temperature ratio parameter's concentration profile is reduced, which is a relief.

The influence of chemical reaction parameter σ on nanoparticle volume fraction profile is illustrated in Supplemental Figure 19 in the case of species formation. The volume fraction of nanoparticles decreases when the constructive chemical reaction parameter is used. This is because the chemical processes in this system consume the chemicals, causing the concentration profile to decrease. The first order chemical reaction has the most important influence in that it tends to decrease overshoot in solute concentration profiles in the solute boundary layer.

The activation energy E 's activities are depicted in Supplemental Figure 20 against a sketch of nanoparticle concentration. The Arrhenius function lowers as the activation energy increases, resulting in a better productive chemical reaction and a higher concentration profile. When you combine greater activation energy with a lesser temperature, you get a lesser reaction rate constant, which condenses the chemical reaction. As a result, the concentration of solutes rises. A considerable quantity of energy is required to start a greater chemical reaction. The Arrhenius function lowers as E increases, and as a result, the chemical action elevations the concentration. By way of increased activation energy and lower temperature, the reaction rate slows down slightly, decelerating the chemical reaction. As a result, the number of species on the planet grows.

Supplemental Figures 21–24 illustrates the contour plots of skin friction with respect to active parameters on magnetic field M , Nr , α , λ , electric field El , and Schmidt number Sc . It is observed from those figures that the skin friction is an extraordinary for the smaller

values of the M , α , Nr and a reverse trend is observed for El , λ , and Sc .

Supplemental Figures 25–28 accords the behavior of the M , Nr , λ , Ec , α , El , and Rd on the rate of heat transfer. From these, Nusselt number has direct relationship with Ec , α , Rd , and Nr . But it has opposite relationship with El and λ . The magnetic field parameter has not much significance on Nusselt number.

Supplemental Figures 29–31 shows the vision of the drove's parameters m , Sc , E , and σ on the Sherwood number. Stronger results in a higher concentration of Sherwood number. However, a declination in Sherwood number Sc is detected for σ . There is no significance impact on M and E on Sherwood number.

Conclusions

In this article, EMHD, multiple slips, and activation energy with Cattaneo–Christov heat flux have been investigated in the context of non-linear radiation generation over a stretched sheet. Correspondence transformations are pragmatic to non-dimensionalize the governing boundary equations. Following that, the partial differential equations are malformed into a system of coupled non-linear ODEs, which are elucidated using the RK fourth order approach with the shooting methodology. For varying values of physical constraints on flow velocity, temperature, concentration, heat and mass transfer rates, the values are graphically shown. The following are the most important implications of the current study:

- When the magnetic field M strength increases, the velocity profile drops, but the temperature profile reverses.
- The velocity drawings rise when the stagnation parameter is increased, but the temperature example shows the opposite trend.
- When the electric field is increased, the skin friction coefficient increases, but when parameters like M , m and Sc are increased, the skin friction coefficient drops.
- By rising the values of El , the temperature profile demolishes.
- Nanoparticle concentration rises for the rise in the parameter E .

Due to their remarkable properties, this theoretical research may be useful for material science, purification technologies, optics, electricity, the aerospace industry, and many other fields. Especially in aerospace, where it may help with weight reduction, satellite weight reduction and aircraft lightning protection, among other things.

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ORCID iD

P. Bala Anki Reddy  <https://orcid.org/0000-0001-5824-1796>

Supplemental material

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